

Econometrics Assignment 3

Applied Time-Series Econometrics

Assignment Overview

This assignment introduces applied time-series econometrics using macroeconomic and financial datasets.

The assignment focuses on:

- stationarity
- trends and seasonality
- autocorrelation
- AR, MA, and ARIMA models
- unit roots
- cointegration
- VAR models
- volatility modeling
- GARCH models
- forecasting
- structural breaks and regime shifts

You are expected to understand both:

- statistical forecasting
- economic interpretation

The assignment emphasizes why non-stationary data creates problems and how modern time-series methods attempt to address those problems.

Topics to Learn Before Attempting This Assignment

A. Foundations of Time Series

- time-series data structure
- trends
- seasonality
- stochastic vs deterministic processes
- white noise
- lag operators
- shocks and persistence

B. Stationarity and Unit Roots

- weak stationarity
- covariance stationarity
- unit roots
- random walks
- differencing
- integrated processes
- spurious regression
- ADF testing

C. Autocorrelation and Dependence

- serial correlation
- ACF
- PACF
- Ljung-Box tests
- persistence
- mean reversion

D. AR, MA, and ARIMA Models

- autoregressive models
- moving average models
- ARMA models
- ARIMA models
- lag selection
- AIC and BIC
- residual diagnostics

E. Forecasting

- train-test splits
- rolling forecasts
- recursive forecasting
- RMSE
- MAE
- forecast comparison
- overfitting in forecasting

F. Cointegration and Long-Run Relationships

- non-stationary regression
- cointegration
- Engle-Granger procedure
- equilibrium relationships
- error correction models

G. Vector Autoregressions

- multivariate time-series systems
- VAR intuition
- lag structure
- impulse response functions
- Granger causality
- forecast error propagation

H. Volatility Models

- conditional heteroskedasticity
- ARCH effects
- volatility clustering
- GARCH models
- persistence of volatility
- financial risk forecasting

I. Structural Breaks and Regime Shifts

- structural breaks
- regime changes
- policy shifts
- crisis periods
- parameter instability
- rolling-window analysis

Required Submission

Submit:

1. One Jupyter notebook
2. One short report

For every problem, include:

- plots
- stationarity tests
- model equations
- estimated coefficients
- diagnostic tests
- forecast metrics
- interpretation

Use:

- statsmodels
- arch
- pandas
- numpy
- matplotlib

Problem 1 — Inflation Dynamics and ARIMA Forecasting

Dataset

Use FRED Consumer Price Index for All Urban Consumers:

CPIAUCSL

Suggested sample:

- Monthly data
- 1980-01 to latest available period

Research Question

Can past inflation help forecast future inflation?

Part A — Construct Inflation

Start with monthly CPI.

Create:

```
df["inflation_m"] =  
100 * np.log(df["CPIAUCSL"]).diff()  
  
df["inflation_ann"] =  
1200 * np.log(df["CPIAUCSL"]).diff()
```

Tasks

1. Plot CPI levels.
2. Plot monthly inflation.
3. Plot annualized monthly inflation.

Questions

1. Which series looks more stationary: CPI or inflation?
2. Why are price levels usually non-stationary?
3. Does inflation appear persistent?
4. Are there visible structural breaks?

Part B — Stationarity and Unit Roots

Run ADF tests on:

- CPI level
- $\log(\text{CPI})$
- monthly inflation
- annualized inflation

Python skeleton:

```
from statsmodels.tsa.stattools import adfuller

result = adfuller(
    df["inflation_ann"].dropna()
)
```

Questions

1. Does CPI have a unit root?
2. Does inflation have a unit root?
3. What does a unit root mean intuitively?
4. Why do economists difference macroeconomic price levels?
5. Why can non-stationary regressions become misleading?

Expected intuition: a unit root means shocks have persistent effects and the series does not naturally return to a fixed mean.

Part C — Autocorrelation

Plot:

- ACF
- PACF

for inflation.

Python skeleton:

```
plot_acf(df["inflation_ann"].dropna(), lags=36)
plot_pacf(df["inflation_ann"].dropna(), lags=36)
```

Questions

1. Is inflation autocorrelated?
2. How many lags appear relevant?
3. Does the PACF suggest AR behavior?
4. Does the ACF suggest MA behavior?

Part D — AR, MA, and ARIMA Models

Estimate:

- AR(1)
- AR(2)
- MA(1)
- ARMA(1,1)
- ARIMA models

Use:

`ARIMA(y, order=(p,d,q))`

Tasks

1. Compare models using:
 - AIC
 - BIC
2. Check residual autocorrelation.
3. Run Ljung-Box tests.

Questions

1. Which model minimizes AIC?
2. Which minimizes BIC?
3. Are AR terms statistically significant?
4. Are MA terms statistically significant?
5. Do residuals still contain autocorrelation?

Part E — Forecasting

Create a train-test split:

- Train: 1980-01 to 2018-12
- Test: 2019-01 onward

Forecast inflation using:

- mean forecast
- AR(1)
- best ARIMA model

Evaluate forecasts using:

- RMSE
- MAE

Questions

1. Which model has the lowest RMSE?
2. Which model has the lowest MAE?
3. Does the more complicated model forecast better?
4. Why can a simple model outperform a complex model?

Problem 2 — Oil Prices, Returns, and Volatility

Dataset

Use FRED WTI crude oil prices:

DCOILWTICO

Suggested sample:

- Daily data
- 2000-01-01 onward

Research Question

Are oil prices predictable, and is oil-price volatility clustered?

Part A — Prices and Returns

Create:

```
df["log_price"] =  
np.log(df["DCOILWTICO"])  
  
df["return"] =  
100 * df["log_price"].diff()
```

Tasks

1. Plot oil prices.
2. Plot daily returns.

Questions

1. Which series appears more stationary?
2. Are volatility clusters visible?
3. Are there structural breaks around crises?

Part B — Unit-Root Testing

Run ADF tests on:

- price
- log price
- returns

Questions

1. Does price appear non-stationary?
2. Do returns appear stationary?
3. Why do financial models usually use returns instead of prices?

Part C — ARIMA on Oil Returns

Estimate:

- AR(1)
- ARMA(1,1)
- ARMA(2,2)

Questions

1. Are returns strongly autocorrelated?
2. Does ARIMA forecast returns well?
3. Are residuals approximately white noise?
4. What does weak return predictability imply?

Part D — ARCH Effects

After estimating a mean model, analyze squared residuals.

Tasks

1. Plot ACF of squared residuals.
2. Run Ljung-Box tests on squared residuals.

Questions

1. Are squared residuals autocorrelated?
2. What does this imply?
3. Why can returns show weak autocorrelation while squared returns show strong autocorrelation?

Expected intuition: returns may be difficult to predict, but volatility may still be predictable.

Part E — GARCH Volatility Model

Estimate:

$$GARCH(1,1)$$

Python skeleton:

```
garch = arch_model(  
    r,  
    mean="Constant",  
    vol="GARCH",  
    p=1,  
    q=1  
) .fit()
```

Questions

1. Is the ARCH term significant?
2. Is the GARCH term significant?
3. Is volatility persistent?
4. Interpret $\alpha + \beta$.
5. Plot conditional volatility.

Part F — Forecast Volatility

Forecast volatility 10 days ahead.

Questions

1. Does volatility revert quickly or slowly?
2. Why does volatility persistence matter for risk management?
3. Why is volatility forecasting easier than return forecasting?

Problem 3 — Inflation, Interest Rates, and Unemployment: VAR Model

Dataset

Use monthly FRED data:

Variable	FRED Series
Unemployment rate	UNRATE
CPI	CPIAUCSL
Federal funds rate	FEDFUNDS

Suggested period:

- 1985-01 onward

Create:

```
df["inflation"] =  
1200 * np.log(df["CPIAUCSL"]).diff()
```

Research Question

How do inflation, unemployment, and interest rates interact dynamically?

Part A — Data Preparation

Tasks

1. Plot all variables.
2. Check persistence visually.
3. Discuss whether differencing is needed.

Questions

1. Which variables appear persistent?
2. Which variables may contain unit roots?
3. Why can estimating VAR in non-stationary levels be dangerous?

Part B — Stationarity Tests

Run ADF tests on:

- inflation
- unemployment
- federal funds rate

Questions

1. Which variables reject the unit-root null?
2. Which do not?
3. When might using levels still be defensible?

Part C — Estimate VAR

Choose lag length using:

- AIC
- BIC

Estimate VAR models.

Questions

1. Which lag length is selected by AIC?
2. Which lag length is selected by BIC?
3. Why does AIC often choose more lags?
4. Are lagged interest rates useful for predicting inflation?

Part D — Granger Causality

Test:

- Does FEDFUNDS Granger-cause inflation?
- Does inflation Granger-cause FEDFUNDS?
- Does unemployment Granger-cause FEDFUNDS?

Questions

1. Which null hypotheses are rejected?
2. Does Granger causality imply true causality?
3. Why or why not?

Important: Granger causality means predictive usefulness, not structural causality.

Part E — Impulse Response Functions

Generate impulse responses.

Questions

1. What happens to inflation after an interest-rate shock?
2. What happens to unemployment after an interest-rate shock?
3. How long do effects persist?
4. Why does variable ordering matter for orthogonalized IRFs?

Part F — Forecasting

Forecast:

- inflation
- unemployment
- federal funds rate

Compare VAR forecasts against univariate AR models.

Questions

1. Does VAR improve inflation forecasting?
2. Does adding unemployment and interest rates help?
3. Compare RMSE and MAE.
4. Why can VAR models overfit?

Problem 4 — Cointegration and Long-Run Equilibrium

Choose one option:

- stock index and ETF
- GDP and consumption

Research Question

Are two non-stationary economic series tied together by a stable long-run relationship?

Part A — Stationarity Analysis

Run ADF tests on:

- levels
- first differences

Questions

1. Are levels non-stationary?
2. Are first differences stationary?
3. Are both variables likely integrated of order one?
4. Why is this important for cointegration analysis?

Part B — Engle-Granger Cointegration Test

Estimate:

$$y_t = \alpha + \beta x_t + u_t$$

Then test whether residuals are stationary.

Questions

1. Are residuals stationary?
2. Are the variables cointegrated?
3. What does cointegration mean intuitively?
4. Why can two non-stationary variables still have a meaningful long-run regression?

Part C — Error Correction Model

Estimate:

$$\Delta y_t = \alpha + \gamma u_{t-1} + \beta_1 \Delta x_t + \epsilon_t$$

Questions

1. Is the error-correction term statistically significant?
2. Is its sign negative?
3. What does adjustment back to equilibrium mean?
4. How fast does the system correct disequilibrium?

Part D — Forecasting Comparison

Forecast using:

- random walk
- ARIMA
- ECM

Questions

1. Which model forecasts best?
2. Does cointegration improve forecasting?
3. Why can long-run equilibrium relationships help medium-term forecasts more than short-term forecasts?

Forecasting Table

Every forecasting exercise must include:

Model	RMSE	MAE	Notes
Mean Forecast			
Random Walk			
ARIMA			
VAR / ECM / GARCH			

Grading Rubric

Area	Marks
Data cleaning and plots	10
Stationarity and unit-root testing	15
Autocorrelation analysis	10
AR/MA/ARIMA estimation	15
Cointegration and ECM	15
VAR modeling and Granger causality	15
GARCH volatility modeling	10
Forecast evaluation	10
Total	100

What You Should Learn

By the end of this assignment, you should understand:

- why stationarity matters
- why non-stationary regressions can fail
- how autocorrelation affects forecasting
- differences between AR, MA, and ARIMA models
- why volatility clustering matters in finance
- how VAR models capture dynamic systems
- why Granger causality is not structural causality
- how cointegration captures long-run equilibrium
- why forecasting is fundamentally difficult
- how structural breaks can destabilize models

Final Reminder

Forecasting models are not automatically causal models.

A model may forecast well while having weak causal interpretation.

Always distinguish:

- prediction
- association
- structural causality